PERSONALISED MODELS OF RESTING-STATE FMRI · ALZHEIMER'S DISEASE

Optimal stimulation sites are not the most affected

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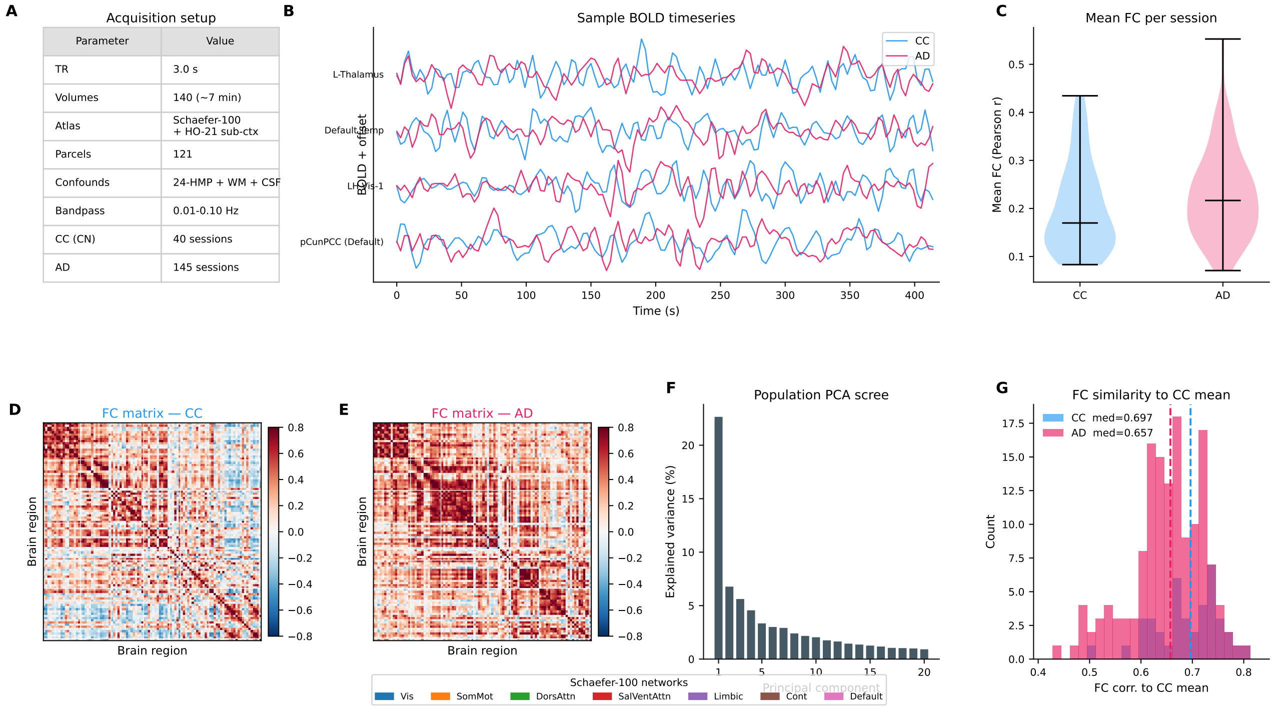
Alzheimer's is a distributed network disorder.

Yet targeted neuromodulation needs a **focal** target.

Where should we stimulate?

Resting-state fMRI: controls and patients

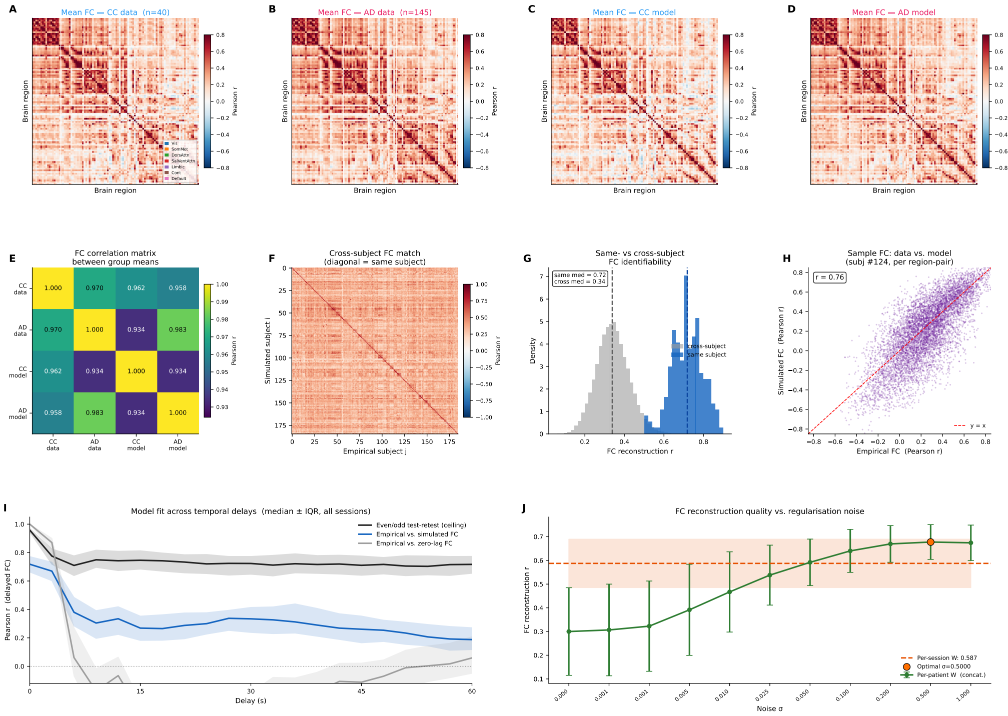


Acquisition setup

Parameter	Value
TR	3.0 s
Volumes	140 (~7 min)
Atlas	Schaefer-100 + HO-21 sub-ctx
Parcels	121
Confounds	24-HMP + WM + CSF
Bandpass	0.01-0.10 Hz
CC (CN)	40 sessions
AD	145 sessions

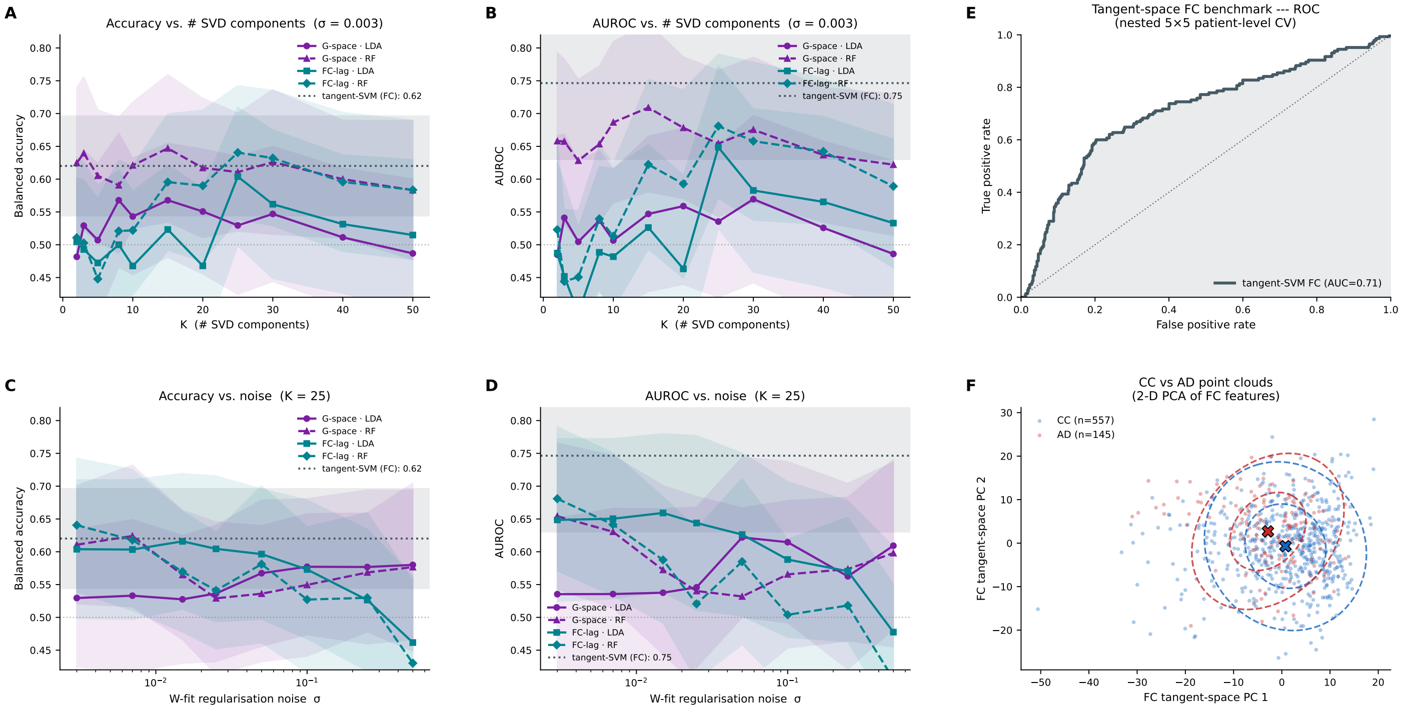
Individual resting-state fMRI from cognitively unimpaired controls and AD patients.

A perturbable digital twin of each patient



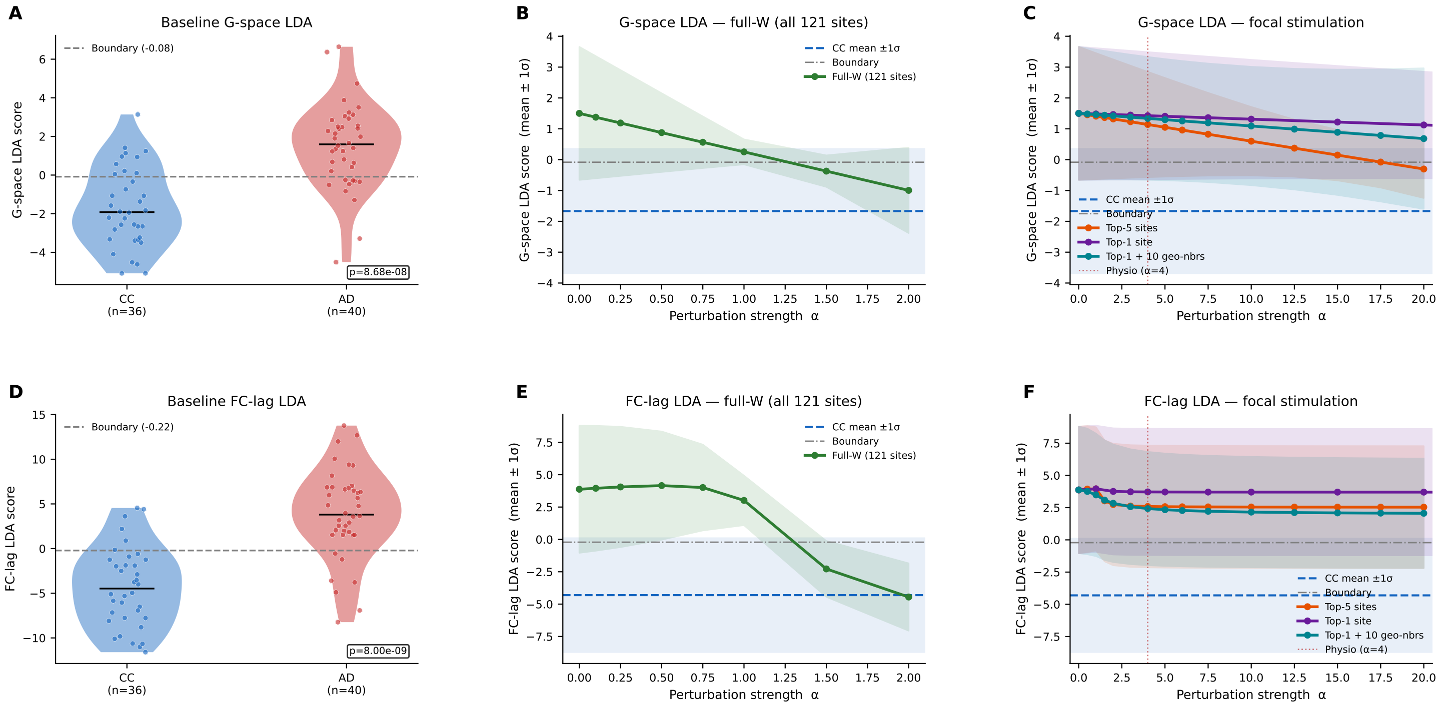
A personalised reservoir model whose free-running dynamics **reproduce the patient's own** resting-state activity.

Reading the disease from the model



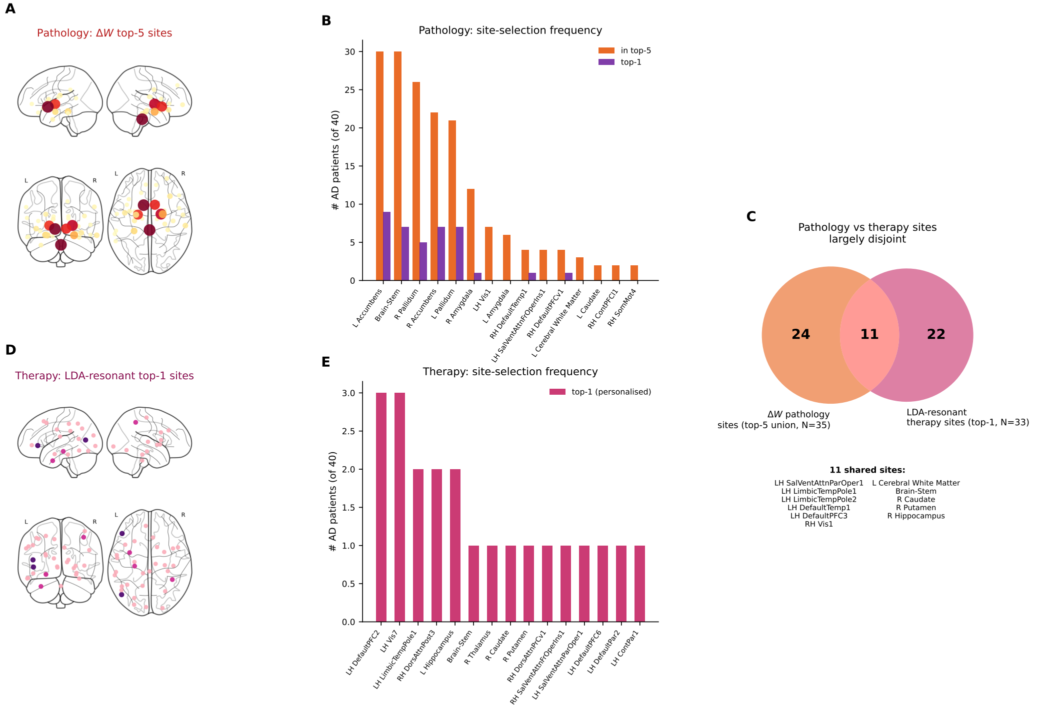
The fitted model parameters separate AD from controls — a *target* for intervention, not a diagnostic end.

Correcting the connectivity reverts the signature



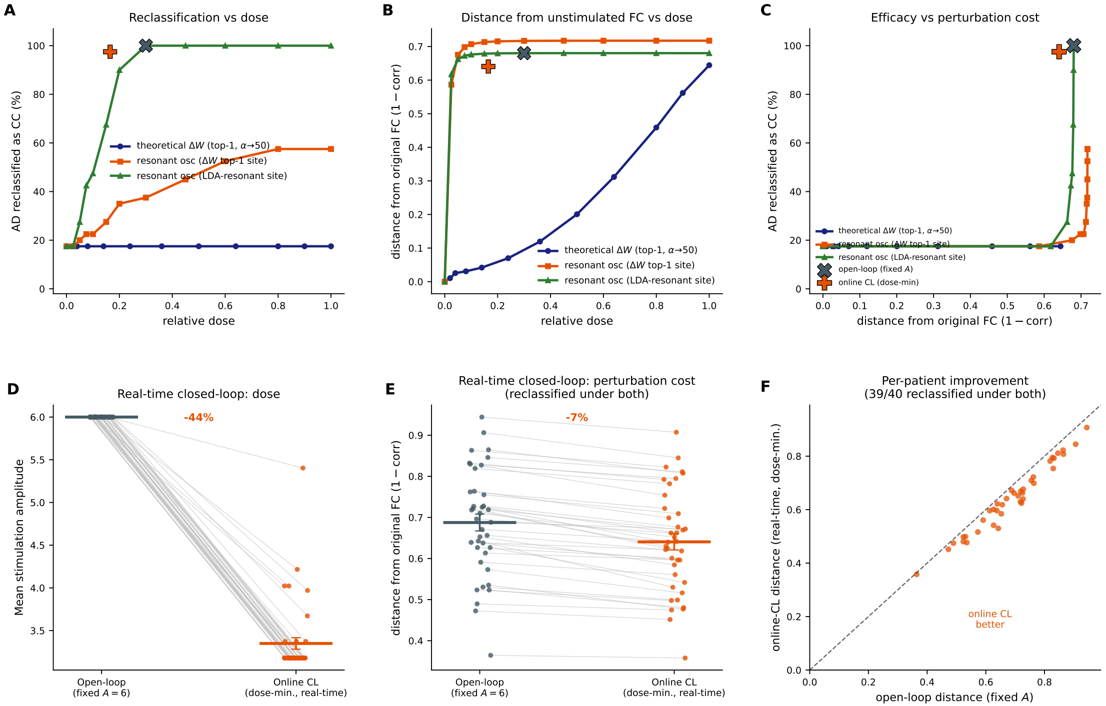
Shifting the whole connectivity toward controls works; focal edits fail at any amplitude — the correction is distributed.

The best target is not the most affected site



Most-changed connectivity sites and effective stimulation targets are largely disjoint.

One site, and closed-loop control



A resonant drive at each patient's discriminant-aligned site reverts the classification; closed-loop titration reaches it at lower dose.

Stimulate where the network responds — not where it is most affected.

Model-informed, personalised targeting turns “where to stimulate” into a computable prescription.